Creating informative maps

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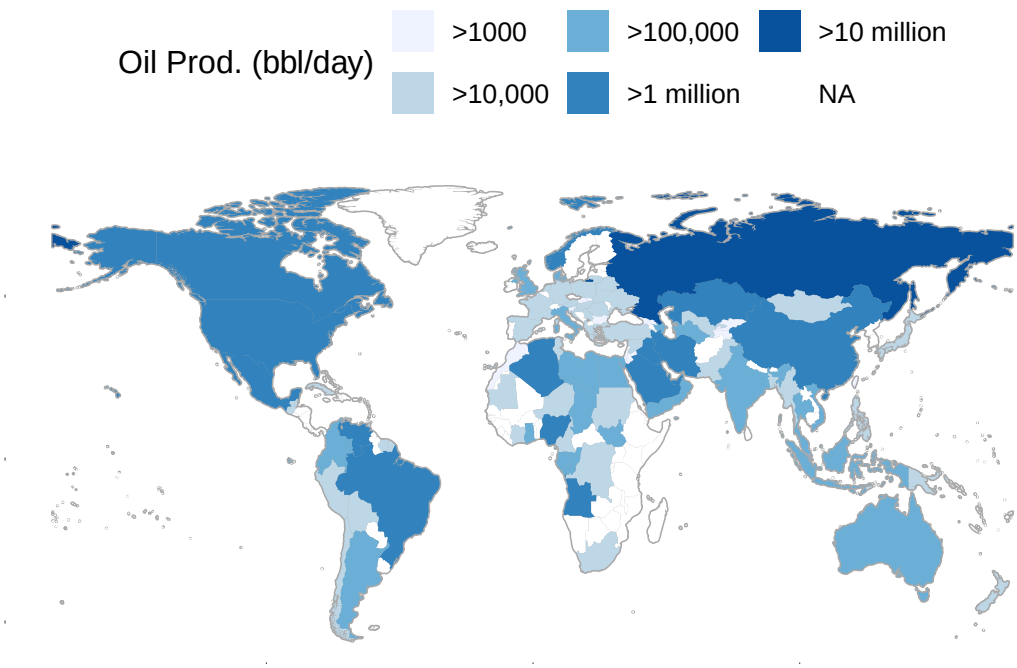
```
# Initial packages required (we'll be adding more)
library(tidyverse)
library(mdsr)      # package associated with our MDSR book
```

Opening example

Here is a simple choropleth map example from Section 3.2.3 of MDSR. Note how we use an underlying map with strategic shading to convey a story about a variable that's been measured on each country.

```
# CIACountries is a 236 x 8 data set with information on each country
#   taken from the CIA factbook - gdp, education, internet use, etc.
head(CIACountries)
CIACountries |>
  select(country, oil_prod) |>
  mutate(oil_prod_disc = cut(oil_prod,
    breaks = c(0, 1e3, 1e5, 1e6, 1e7, 1e8),
    labels = c(">1000", ">10,000", ">100,000", ">1 million",
               ">10 million"))) |>
  mosaic::mWorldMap(key = "country") +                                    ①
  geom_polygon(aes(fill = oil_prod_disc)) +
  scale_fill_brewer("Oil Prod. (bbl/day)", na.value = "white") +
  theme(legend.position = "top")
```

① We won't use mWorldMap often, but it's a good quick illustration




```
         country      pop    area oil_prod   gdp educ   roadways net_users
1    Afghanistan 32564342  652230        0  1900   NA 0.06462444       >5%
2        Albania  3029278   28748    20510 11900  3.3 0.62613051      >35%
3        Algeria 39542166 2381741  1420000 14500  4.3 0.04771929      >15%
4 American Samoa    54343     199        0 13000   NA 1.21105528      <NA>
5        Andorra    85580     468       NA 37200   NA 0.68376068      >60%
6         Angola 19625353 1246700  1742000  7300  3.5 0.04125211      >15%
```

Choropleth Maps

When you have specific regions (e.g. countries, states, counties, census tracts,...) and a value associated with each region.

A choropleth map will color the entire region according to the value. For example, let's consider state vaccination data from March 2021.

```
vaccines <- read_csv("https://joeroith.github.io/264_spring_2026/Data/vacc_Mar21.csv")

vacc_mar13 <- vaccines |>
  filter(Date =="2021-03-13") |>
  select(State, Date, people_vaccinated_per100, share_doses_used, Governor)

vacc_mar13
```

```
# A tibble: 50 x 5
   State       Date       people_vaccinated_per100 share_doses_used Governor
   <chr>       <date>                        <dbl>            <dbl> <chr>
 1 Alabama     2021-03-13                     17.2            0.671 R
 2 Alaska      2021-03-13                     27.0            0.686 R
 3 Arizona     2021-03-13                     21.5            0.821 R
 4 Arkansas    2021-03-13                     19.2            0.705 R
 5 California  2021-03-13                     20.3            0.726 D
 6 Colorado    2021-03-13                     20.8            0.801 D
 7 Connecticut 2021-03-13                     26.2            0.851 D
 8 Delaware    2021-03-13                     20.2            0.753 D
 9 Florida     2021-03-13                     20.1            0.766 R
10 Georgia     2021-03-13                     15.2            0.674 R
# i 40 more rows
```

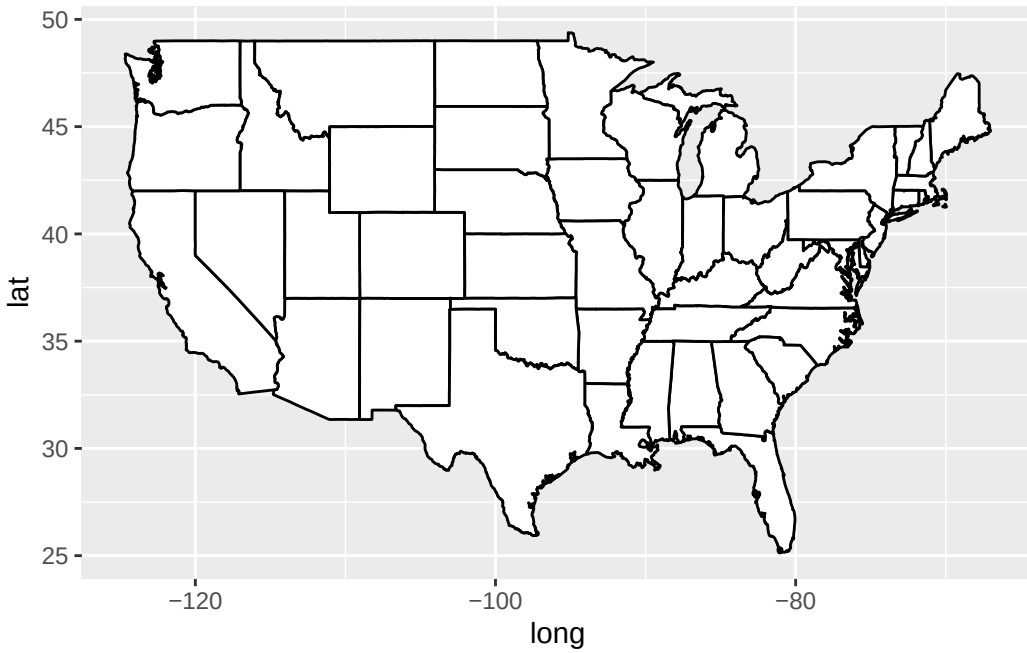
The tricky part of choropleth maps is getting the shapes (polygons) that make up the regions. This is really a pretty complex set of lines for R to draw!

Luckily, some maps are already created in R in the `maps` package.

```r
library(maps)
us_states <- map_data("state")
head(us_states)
```

```
       long      lat group order  region subregion
1 -87.46201 30.38968     1     1 alabama      <NA>
2 -87.48493 30.37249     1     2 alabama      <NA>
3 -87.52503 30.37249     1     3 alabama      <NA>
4 -87.53076 30.33239     1     4 alabama      <NA>
5 -87.57087 30.32665     1     5 alabama      <NA>
6 -87.58806 30.32665     1     6 alabama      <NA>
```

```r
us_states |>
  ggplot(mapping = aes(x = long, y = lat,
                       group = group)) +
  geom_polygon(fill = "white", color = "black")
```



[**Pause to ponder:**] What might the `group` and `order` columns represent? *Group tells R which point belong to the same shape, not just state but also subregion as well. 'order' looks like it ranks the latitude from highest to lowest. Oreder is also about what drawing order within that shape*

Other maps provided by the `maps` package include US counties, France, Italy, New Zealand, and two different views of the world. If you want maps of other countries or regions, you can often find them online.

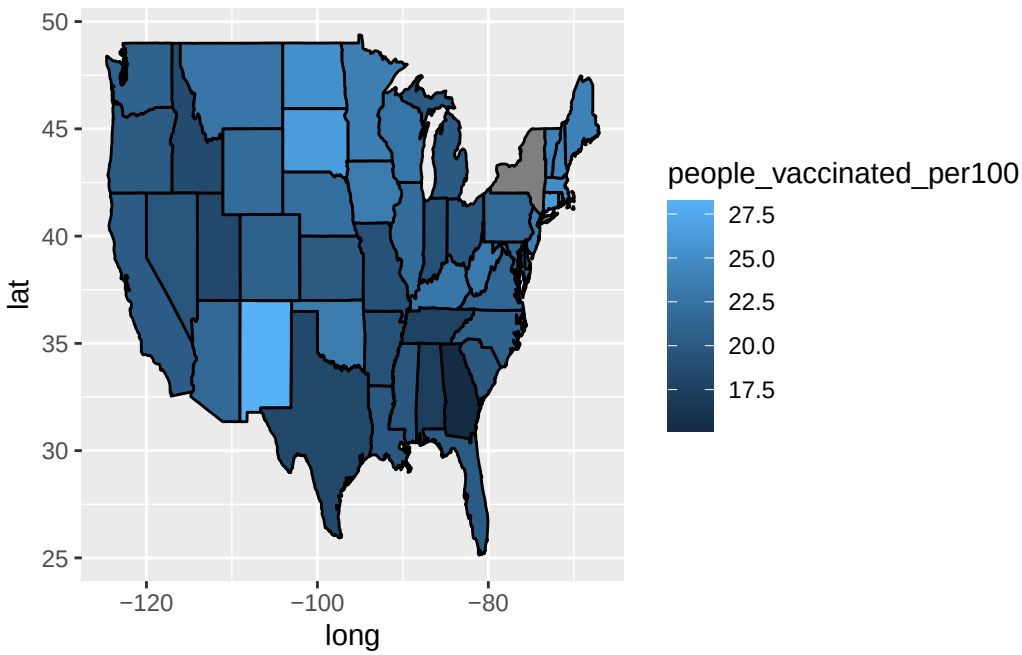
Where the really cool stuff happens is when we join our data to the `us_states` dataframe. Notice that the state name appears in the "region" column of `us_states`, and that the state name is in all small letters. In `vacc_mar13`, the state name appears in the State column and is capitalized. Thus, we have to be very careful when we join the state vaccine info to the state geography data.

Run this line by line to see what it does:

```
vacc_mar13 <- vacc_mar13 |>
  mutate(State = str_to_lower(State))

vacc_mar13 |>
  right_join(us_states, by = c("State" = "region")) |>
  rename(region = State) |>
  ggplot(mapping = aes(x = long, y = lat,
```

```
                                group = group)) +
  geom_polygon(aes(fill = people_vaccinated_per100), color = "black")
```



oops, New York appears to be a problem.

```
vacc_mar13 |>
  anti_join(us_states, by = c("State" = "region"))
```

```
# A tibble: 3 x 5
  State          Date       people_vaccinated_per100 share_doses_used Governor
  <chr>          <date>                        <dbl>            <dbl> <chr>
1 alaska         2021-03-13                     27.0            0.686 R
2 hawaii         2021-03-13                     22.8            0.759 D
3 new york state 2021-03-13                     21.7            0.764 D
```

```
us_states |>
  anti_join(vacc_mar13, by = c("region" = "State")) |>
  count(region)
```

```
                region   n
1 district of columbia  10
2             new york 495
```

[**Pause to ponder:**] What did we learn by running `anti_join()` above? *Shows the mismatched of datasets (the first data set is New york and the second one is New york state for example)*

Notice that the `us_states` map also includes only the contiguous 48 states. This gives an example of creating really beautiful map insets for Alaska and Hawaii.

```
vacc_mar13 <- vacc_mar13 |>
  mutate(State = str_replace(State, " state", ""))

vacc_mar13 |>
  anti_join(us_states, by = c("State" = "region"))
```

```
# A tibble: 2 x 5
  State  Date       people_vaccinated_per100 share_doses_used Governor
  <chr>  <date>                        <dbl>            <dbl> <chr>
1 alaska 2021-03-13                     27.0            0.686 R
2 hawaii 2021-03-13                     22.8            0.759 D
```

```
us_states |>
  anti_join(vacc_mar13, by = c("region" = "State")) %>%
  count(region)
```

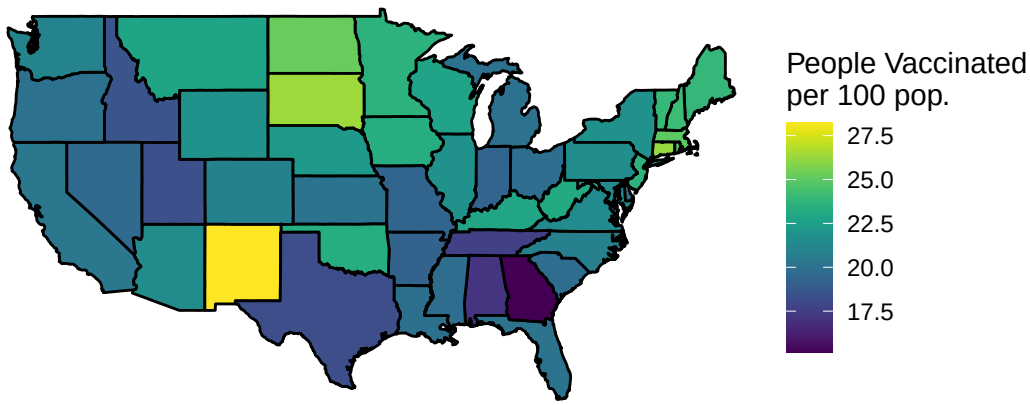
```
                region  n
1 district of columbia 10
```

Better.

```
library(viridis) # for color schemes
vacc_mar13 |>
  right_join(us_states, by = c("State" = "region")) |>
  rename(region = State) |>
  ggplot(mapping = aes(x = long, y = lat,
                       group = group)) +
  geom_polygon(aes(fill = people_vaccinated_per100), color = "black") +
  labs(fill = "People Vaccinated\nper 100 pop.") +
  coord_map() +                                                          ①
  theme_void() +                                                         ②
  scale_fill_viridis()                                                   ③
```

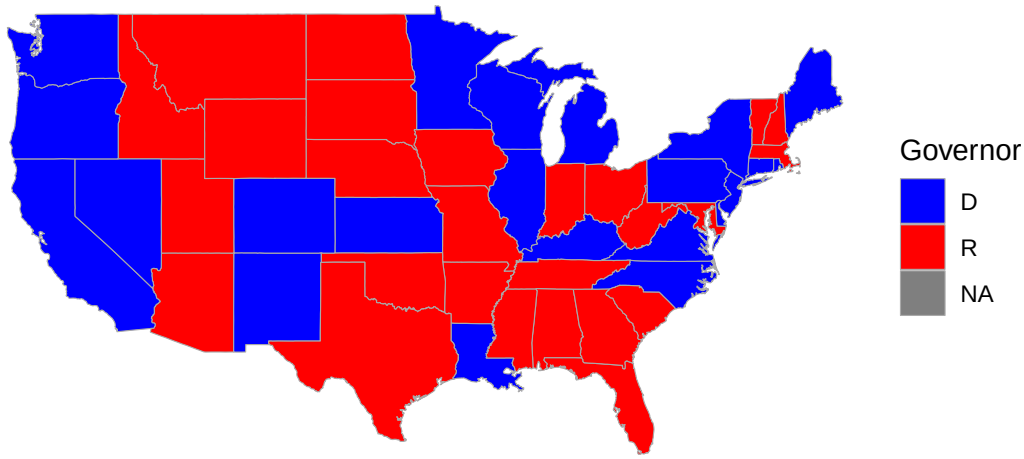
① This scales the longitude and latitude so that the shapes look correct. coord_quickmap() can also work here - it's less exact but faster.

② This theme can give you a really clean look
③ You can change the fill scale for different color schemes.



You can also use a categorical variable to color regions:

```
vacc_mar13 |>
  right_join(us_states, by = c("State" = "region")) |>
  rename(region = State) |>
  ggplot(mapping = aes(x = long, y = lat,
                       group = group)) +
  geom_polygon(aes(fill = Governor), color = "darkgrey", linewidth = 0.2) +
  labs(fill = "Governor") + # this is where it changes
  coord_map() +
  theme_void() +
  scale_fill_manual(values = c("blue", "red")) #<1> #manually fill blue and red
```



1. You can change the fill scale for different color schemes.

Note: Map projections are actually pretty complicated, especially if you're looking at large areas (e.g. world maps) or drilling down to very small regions where a few feet can make a difference (e.g. tracking a car on a map of roads). It's impossible to preserve both shape and area when projecting an (imperfect) sphere onto a flat surface, so that's why you sometimes see such different maps of the world. This is why packages like `maps` which connect latitude-longitude points are being phased out in favor of packages like `sf` with more GIS functionality. We won't get too deep into GIS in this class, but to learn more, take Spatial Data Analysis!!

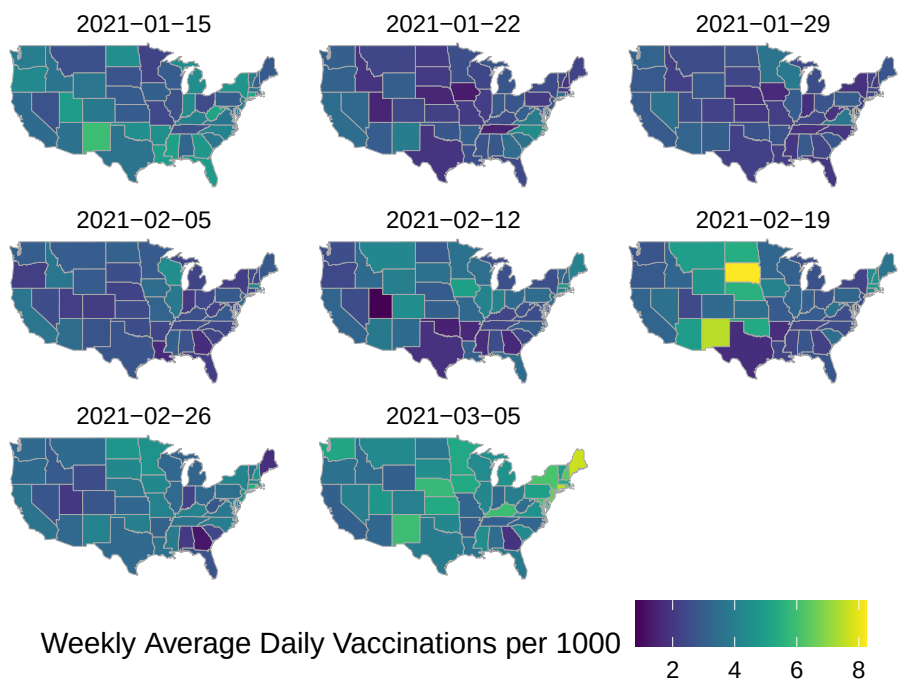
Multiple maps!

You can still use data viz tools from Data Science 1 (like faceting) to create things like time trends in maps:

```
library(viridis)
library(lubridate)
weekly_vacc <- vaccines |>
  mutate(State = str_to_lower(State)) |>
  mutate(State = str_replace(State, " state", ""),
         week = week(Date)) |>
  group_by(week, State) |>
  summarize(date = first(Date),
```

```
            mean_daily_vacc = mean(daily_vaccinated/est_population*1000)) |>
  right_join(us_states, by =c("State" = "region")) |>
  rename(region = State)

weekly_vacc |>
  filter(week > 2, week < 11) |>
  ggplot(mapping = aes(x = long, y = lat,
                       group = group)) +
  geom_polygon(aes(fill = mean_daily_vacc), color = "darkgrey",
               linewidth = 0.1) +
  labs(fill = "Weekly Average Daily Vaccinations per 1000") +
  coord_map() +
  theme_void() +
  scale_fill_viridis() +
  facet_wrap(~date) +
  theme(legend.position = "bottom")
```



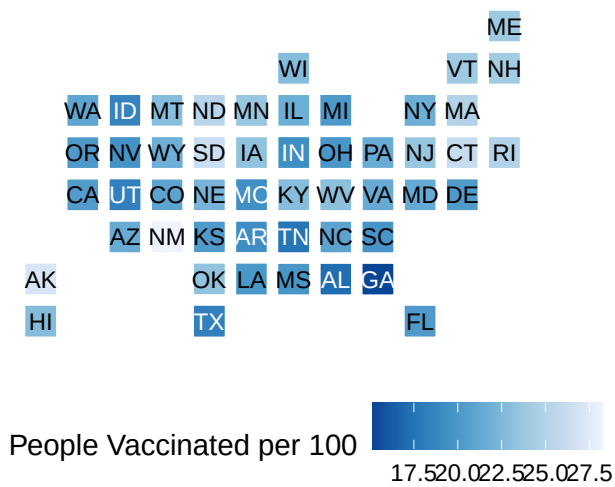
[**Pause to ponder:**] are we bothered by the warning about many-to-many when you run the code above? *Not really. the map data has many rows per state becayse each states is made of many boundary points.The weekly summary has one row per state-week. So after the join, each state-week value gets repeated across all the polygon points for that state. That is expected for map drawing.*

Other cool state maps

statebin (square representation of states)

```
library(statebins) # may need to install

vacc_mar13 |>
  mutate(State = str_to_title(State)) |>
  statebins(state_col = "State",
            value_col = "people_vaccinated_per100") + # what put the data into layout, get r
  theme_statebins() + #<1> # position them into the actaul shape of
  labs(fill = "People Vaccinated per 100")
```



1. One nice layout. You can customize with usual ggplot themes.

[**Pause to ponder:**] Why might one use a map like above instead of our previous choropleth maps? *Better comparison and might also because they all have the same square size*

I used this example to create the code above. The original graph is located here.

Interactive point maps with leaflet

To add even more power and value to your plots, we can add interactivity. For now, we will use the `leaflet` package, but later in the course we will learn even more powerful and flexible approaches for creating interactive plots and webpages.

For instance, here is a really simple plot with a pop-up window:

```
library(leaflet)

leaflet() |>
  addTiles() |>   #<1> # add the background
  setView(-93.1832, 44.4597, zoom = 17) |>                                  ②
  addPopups(-93.1832, 44.4597, 'Here is the <b>Regents Hall of Mathematical Sciences</b>, hor
```

② setView() centers the map at a specific latitude and longitude, then zoom controls how much of the surrounding area is shown
③ add a popup message (with html formatting) that can be clicked on or off

Leaflet is not part of the tidyverse, but the structure of its code is pretty similar and it also plays well with piping.

Let's try pop-up messages with a data set containing Airbnb listings in the Boston area:

```
leaflet() |>
    addTiles() |>
    setView(lng = mean(airbnb.df$Long), lat = mean(airbnb.df$Lat),
            zoom = 13) |>
    addCircleMarkers(data = airbnb.df,  #add circle
        lat = ~ Lat,
        lng = ~ Long,
        popup = ~ AboutListing, #pop up include the list
        radius = ~ S_Accomodates,
        # These last options describe how the circles look
        weight = 2,
        color = "red",
        fillColor = "yellow")
```

[**Pause to ponder:**] List similarities and differences between leaflet plots and ggplots. *Static = map ggplot, each boarder point gets its own row, long dataset , cannot use this to make interactive plot interactive - leaflet, sf, can uses these to make static plot - ggplot2 is for static plots, and slo often uses plain coordinate columns or sf - Leaflet is for interactive, useful for zooming, clicking ect*

Interactive choropleth maps with leaflet

OK. Now let's see if we can put things together and duplicate the interactive choropleth map found here showing population density by state in the US.

A preview to shapefiles and the sf package

```
library(sf)                                                                  ①
states <- read_sf("https://rstudio.github.io/leaflet/json/us-states.geojson") ②
class(states)                                                                ③
states
```

① `sf` stands for "simple features"
② From https://leafletjs.com/examples/choropleth/us-states.js
③ Note that `states` has class `sf` in addition to the usual `tbl` and `df`

```
[1] "sf"         "tbl_df"     "tbl"        "data.frame"
Simple feature collection with 52 features and 3 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -188.9049 ymin: 17.92956 xmax: -65.6268 ymax: 71.35163
Geodetic CRS:  WGS 84
# A tibble: 52 x 4
   id    name                  density                                  geometry
   <chr> <chr>                   <dbl>                         <MULTIPOLYGON [°]>
 1 01    Alabama                 94.6  (((-87.3593 35.00118, -85.60667 34.98475~
 2 02    Alaska                   1.26 (((-131.602 55.11798, -131.5692 55.28229~
 3 04    Arizona                 57.0  (((-109.0425 37.00026, -109.048 31.33163~
 4 05    Arkansas                56.4  (((-94.47384 36.50186, -90.15254 36.4963~
 5 06    California             242.   (((-123.2333 42.00619, -122.3789 42.0116~
 6 08    Colorado                49.3  (((-107.9197 41.00391, -105.729 40.99843~
 7 09    Connecticut            739.   (((-73.05353 42.03905, -71.79931 42.0226~
 8 10    Delaware               464.   (((-75.41409 39.80446, -75.5072 39.68396~
 9 11    District of Columbia 10065    (((-77.03526 38.99387, -76.90929 38.8952~
10 12    Florida                353.   (((-85.49714 30.99754, -85.00421 31.0030~
# i 42 more rows
```

For maps in `leaflet` that show boundaries and not just points, we need to input a shapefile rather than a series of latitude-longitude combinations like we did for the `maps` package. In the example we're emulating, they use the `read_sf()` function from the `sf` package to read in data. While our `us_states` data frame from the `maps` package contained 15537 rows, our simple

features object `states` contains only 52 rows - one per state. Importantly, `states` contains a column called `geometry`, which is a "multipolygon" with all the information necessary to draw a specific state. Also, while `states` can be treated as a tibble or data frame, it is also an `sf` class object with a specific "geodetic coordinate reference system". Again, take Spatial Data Analysis for more on shapefiles and simple features!

Note also that the authors of this example have already merged state population densities with state geometries, but if we wanted to merge in other state characteristics using the `name` column as a key, we could definitely do this!

First we'll start with a static plot using a simple features object and geom_sf():

```
# Create density bins as on the webpage
state_plotting_sf <- states |>
  mutate(density_intervals = cut(density, n = 8,  # cut(density..) bins population density in
          breaks = c(0, 10, 20, 50, 100, 200, 500, 1000, Inf))) |> #beak() how you want to cu
  filter(!(name %in% c("Alaska", "Hawaii", "Puerto Rico")))

ggplot(data = state_plotting_sf) +
  geom_sf(aes(fill = density_intervals), colour = "white", linetype = 2) +
#  geom_sf_label(aes(label = density)) +   # labels too busy here
  theme_void() +
  scale_fill_brewer(palette = "YlOrRd")
```

Now let's use `leaflet` to create an interactive plot!

```
# Create our own category bins for population densities
#   and assign the yellow-orange-red color palette
bins <- c(0, 10, 20, 50, 100, 200, 500, 1000, Inf)
pal <- colorBin("YlOrRd", domain = states$density, bins = bins)

# Create labels that pop up when we hover over a state.  The labels must
#   be part of a list where each entry is tagged as HTML code.
library(htmltools)
library(glue)

states <- states |>
  mutate(labels = str_c(name, ": ", density, " people / sq mile"))

# If want more HTML formatting, use these lines instead of those above:
#states <- states |>
#  mutate(labels = glue("<strong>{name}</strong><br/>{density} people / #mi<sup>2</sup>"))
```

```r
labels <- lapply(states$labels, HTML)

leaflet(states) |>
  setView(-96, 37.8, 4) |>
  addTiles() |>
  addPolygons(
    fillColor = ~pal(density),
    weight = 2,
    opacity = 1,
    color = "white",
    dashArray = "3",
    fillOpacity = 0.7,
    highlightOptions = highlightOptions(
      weight = 5,
      color = "#666",
      dashArray = "",
      fillOpacity = 0.7,
      bringToFront = TRUE),
    label = labels,
    labelOptions = labelOptions(
      style = list("font-weight" = "normal", padding = "3px 8px"),
      textsize = "15px",
      direction = "auto")) |>
  addLegend(pal = pal, values = ~density, opacity = 0.7, title = NULL,
    position = "bottomright")
```

[**Pause to ponder:**] Pick several formatting options in the code above, determine what they do, and then change them to create a customized look.

On Your Own

1. Let's return to our example from above where we recreated an interactive choropleth map of population densities by US state. Recall how that plot was very suspicious? The population density data that came with the state geometries from our source seemed incorrect.

In `09_table_scraping.qmd` (On Your Own #1) we went out and found the real statewise population densities (from here) and created a tidy data frame. Now let's merge that with our state geometry shapefiles and regenerate our interactive choropleth map.

```r
# Code from OYO #1 in 09_table_scraping.qmd

library(rvest)
```

```
Attaching package: 'rvest'

The following object is masked from 'package:readr':

    guess_encoding
```

```r
library(polite)

session <- bow("https://en.wikipedia.org/wiki/List_of_states_and_territories_of_the_United_St

result <- scrape(session) |>
  html_nodes(css = "table") |>
  html_table(header = TRUE, fill = TRUE)
density_table <- result[[1]]
density_table
```

```
# A tibble: 61 x 6
   Location               Density Density Population `Land area` `Land area`
   <chr>                  <chr>   <chr>   <chr>      <chr>       <chr>
 1 Location               /mi2    /km2    Population mi2         km2
 2 District of Columbia   11,131  4,297   678,972    61          158
 3 New Jersey             1,263   488     9,290,842  7,354       19,047
 4 Rhode Island           1,060   409     1,095,962  1,034       2,678
 5 Puerto Rico            936     361     3,205,691  3,424       8,868
 6 Massachusetts          898     347     7,001,399  7,800       20,202
 7 Guam[4]                824     319     172,952    210         543
 8 Connecticut            747     288     3,617,176  4,842       12,542
 9 U.S. Virgin Islands[4] 737     284     98,750     134         348
10 Maryland               637     246     6,180,253  9,707       25,142
# i 51 more rows
```

```r
density_data <- density_table |>
  select(1, 2, 4, 5) |>
  filter(!row_number() == 1) |>
  rename(Land_area = `Land area`) |>
```

```r
  mutate(state_name = str_to_lower(as.character(Location)),
         Density = parse_number(Density),
         Population = parse_number(Population),
         Land_area = parse_number(Land_area)) |>
  select(-Location)

density_data
```

```
# A tibble: 60 x 4
   Density Population Land_area state_name
     <dbl>      <dbl>     <dbl> <chr>
 1   11131     678972        61 district of columbia
 2    1263    9290842      7354 new jersey
 3    1060    1095962      1034 rhode island
 4     936    3205691      3424 puerto rico
 5     898    7001399      7800 massachusetts
 6     824     172952       210 guam[4]
 7     747    3617176      4842 connecticut
 8     737      98750       134 u.s. virgin islands[4]
 9     637    6180253      9707 maryland
10     578      43915        76 american samoa[4]
# i 50 more rows
```

```r
# merge with the shapefile
state1 <- read_sf("https://rstudio.github.io/leaflet/json/us-states.geojson") |>
  select(name, geometry)|>
  mutate(state_name = str_to_lower(name))
density_map_sf <- state1 |>
  left_join(density_data, by = "state_name")|>
  filter(!(name %in% c( "Alaska", " Hawaii", "Puerto Rico")))

# Now I want to check
density_map_sf |>
  st_drop_geometry() |>
  count()
```

```
# A tibble: 1 x 1
      n
  <int>
1    50
```

```r
# Making interactive map
# Create color bins
bins <- c(0, 10, 25, 50, 100, 250, 500, 1000, Inf)
pal <- colorBin("YlOrRd", domain = density_map_sf$Density, bins = bins)

# Create hover labels
labels <- density_map_sf |>
  mutate(
    label_text = str_c(
      "<strong>", name, "</strong><br/>",
      "Density: ", round(Density, 1), " people / sq mile<br/>",
      "Population: ", scales::comma(Population), "<br/>",
      "Land area: ", scales::comma(Land_area), " sq mi"
    )
  ) |>
  pull(label_text) |>
  lapply(HTML)

leaflet(density_map_sf) |>
  addTiles() |>
  setView(-96, 37.8, 4) |>
  addPolygons(
    fillColor = ~pal(Density),
    weight = 1,
    color = "black",
    opacity = 1,
    fillOpacity = 0.8,
    highlightOptions = highlightOptions(
      weight = 3,
      color = "#888",
      bringToFront = TRUE
    ),
    label = labels,
    labelOptions = labelOptions(
      direction = "auto",
      textsize = "14px",
      style = list("font-weight" = "normal", padding = "3px 8px")
    )
  ) |>
  addLegend(
    pal = pal,
    values = ~Density,
```

```
    opacity = 0.8,
    title = "Population density",
    position = "bottomright"
  )
```

2. The `states` dataset in the `poliscidata` package contains 135 variables on each of the 50 US states. See here for more detail.

Your task is to create a *two* meaningful choropleth plots, one using a numeric variable and one using a categorical variable from `poliscidata::states`. You should make *two* versions of each plot: a static plot using the `maps` package and `ggplot()`, and an interactive plot using the `sf` package and `leaflet()`. Write a sentence or two describing what you can learn from each plot.

Here's some R code and hints to get you going:

```
# Get info to draw US states for geom_polygon (connect the lat-long points)
library(maps)
library(poliscidata)

states_polygon <- as_tibble(map_data("state")) |>
  select(region, group, order, lat, long)

# See what the state (region) levels look like in states_polygon
unique(states_polygon$region)
```

```
 [1] "alabama"              "arizona"              "arkansas"
 [4] "california"           "colorado"             "connecticut"
 [7] "delaware"             "district of columbia" "florida"
[10] "georgia"              "idaho"                "illinois"
[13] "indiana"              "iowa"                 "kansas"
[16] "kentucky"             "louisiana"            "maine"
[19] "maryland"             "massachusetts"        "michigan"
[22] "minnesota"            "mississippi"          "missouri"
[25] "montana"              "nebraska"             "nevada"
[28] "new hampshire"        "new jersey"           "new mexico"
[31] "new york"             "north carolina"       "north dakota"
[34] "ohio"                 "oklahoma"             "oregon"
[37] "pennsylvania"         "rhode island"         "south carolina"
[40] "south dakota"         "tennessee"            "texas"
[43] "utah"                 "vermont"              "virginia"
[46] "washington"           "west virginia"        "wisconsin"
[49] "wyoming"
```

```r
# Get info to draw US states for geom_sf and leaflet (simple features object
#   with multipolygon geometry column)
library(sf)
states_sf <-
  read_sf("https://rstudio.github.io/leaflet/json/us-states.geojson") |>
  select(name, geometry)

# See what the state (name) levels look like in states_sf
unique(states_sf$name)
```

```
 [1] "Alabama"              "Alaska"               "Arizona"
 [4] "Arkansas"             "California"           "Colorado"
 [7] "Connecticut"          "Delaware"             "District of Columbia"
[10] "Florida"              "Georgia"              "Hawaii"
[13] "Idaho"                "Illinois"             "Indiana"
[16] "Iowa"                 "Kansas"               "Kentucky"
[19] "Louisiana"            "Maine"                "Maryland"
[22] "Massachusetts"        "Michigan"             "Minnesota"
[25] "Mississippi"          "Missouri"             "Montana"
[28] "Nebraska"             "Nevada"               "New Hampshire"
[31] "New Jersey"           "New Mexico"           "New York"
[34] "North Carolina"       "North Dakota"         "Ohio"
[37] "Oklahoma"             "Oregon"               "Pennsylvania"
[40] "Rhode Island"         "South Carolina"       "South Dakota"
[43] "Tennessee"            "Texas"                "Utah"
[46] "Vermont"              "Virginia"             "Washington"
[49] "West Virginia"        "Wisconsin"            "Wyoming"
[52] "Puerto Rico"
```

```r
# Load in state-wise data for filling our choropleth maps
#   (Note that I selected my two variables of interest to simplify)
library(poliscidata)   # may have to install first
polisci_data <- as_tibble(poliscidata::states) |>
  select(state, carfatal07, cook_index3)

# See what the state (state) levels look like in polisci_data
unique(polisci_data$state)   # can't see trailing spaces but can see
```

```
 [1] Alaska
 [2] Alabama
 [3] Arkansas
```

```
 [4] Arizona
 [5] California
 [6] Colorado
 [7] Connecticut
 [8] Delaware
 [9] Florida
[10] Georgia
[11] Hawaii
[12] Iowa
[13] Idaho
[14] Illinois
[15] Indiana
[16] Kansas
[17] Kentucky
[18] Louisiana
[19] Massachusetts
[20] Maryland
[21] Maine
[22] Michigan
[23] Minnesota
[24] Missouri
[25] Mississippi
[26] Montana
[27] NorthCarolina
[28] NorthDakota
[29] Nebraska
[30] NewHampshire
[31] NewJersey
[32] NewMexico
[33] Nevada
[34] NewYork
[35] Ohio
[36] Oklahoma
[37] Oregon
[38] Pennsylvania
[39] RhodeIsland
[40] SouthCarolina
[41] SouthDakota
[42] Tennessee
[43] Texas
[44] Utah
[45] Virginia
[46] Vermont
```

```
[47] Washington
[48] Wisconsin
[49] WestVirginia
[50] Wyoming
50 Levels: Alabama                                        ...
```

```r
                                #   lack of internal spaces
print(polisci_data)   # can see trailing spaces
```

```
# A tibble: 50 x 3
   state                                          carfatal07 cook_index3
   <fct>                                               <dbl> <fct>
 1 "Alaska                                     "       15.2 More Rep
 2 "Alabama                                    "       25.9 More Rep
 3 "Arkansas                                   "       23.7 More Rep
 4 "Arizona                                    "       17.6 Even
 5 "California                                 "       11.7 More Dem
 6 "Colorado                                   "       12.3 Even
 7 "Connecticut                                "        8.7 More Dem
 8 "Delaware                                   "       13.6 More Dem
 9 "Florida                                    "       18.1 Even
10 "Georgia                                    "       18.5 Even
# i 40 more rows
```

R code hints:

- stringr functions like `str_squish` and `str_to_lower` and `str_replace_all` (be sure to carefully look at your keys!)
- *_join functions (make sure they preserve classes)
- filter so that you only have 48 contiguous states (and maybe DC)
- for help with colors: https://rstudio.github.io/leaflet/reference/colorNumeric.html
- be sure labels pop up when scrolling with leaflet

```r
# Make sure all keys have the same format before joining:
#   all lower case, no internal or external spaces
 polisci_data1 <- polisci_data |>
  mutate(
    state_key = state |>
      str_squish() |>
      str_to_lower() |>
      str_replace_all(" ", "")
  )
```

```r
states_polygon1 <- states_polygon |>
  mutate(
    state_key = region |>
      str_squish() |>
      str_to_lower() |>
      str_replace_all(" ", "")
  )

states_sf1 <- states_sf |>
  mutate(
    state_key = name |>
      str_squish() |>
      str_to_lower() |>
      str_replace_all(" ", "")
  )
```

```r
# Now we can merge data sets together for the static and the interactive plots


# Merge with states_polygon (static)

states_static <- states_polygon1 |>
  left_join(polisci_data1, by = "state_key") |>
  filter(!(region %in% c( "Alaska", " Hawaii", "Puerto Rico")))
# Check that merge worked for 48 contiguous states
states_static |>
  distinct(region)|>
  count()
```

```
# A tibble: 1 x 1
      n
  <int>
1    49
```

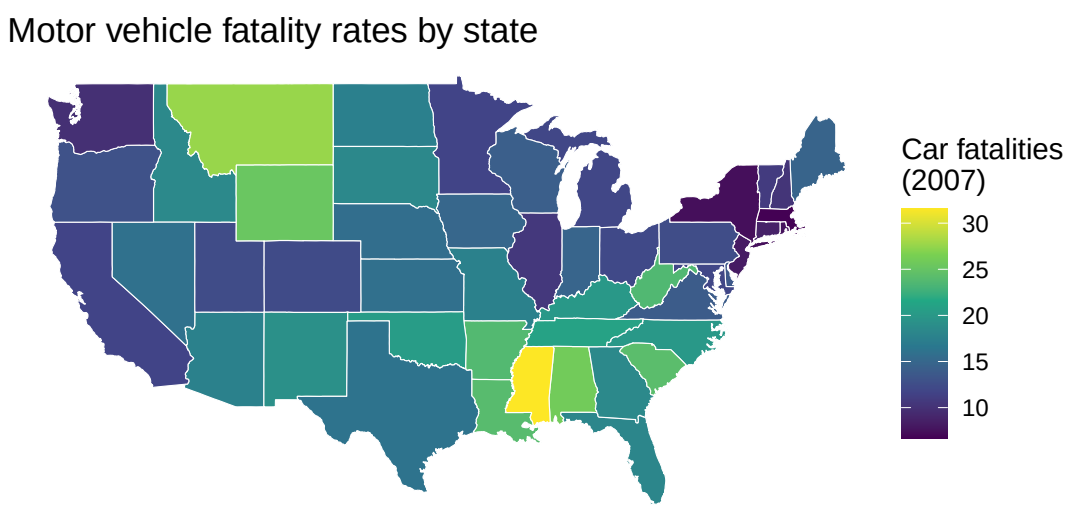
```r
# Merge with states_sf (static or interactive)
states_interactive <-states_sf1 |>
  left_join(polisci_data1, by = "state_key") |>
  filter(!( name %in% c("Alaska", " Hawaii", " Puerto Rico")))
# Check that merge worked for 48 contiguous states
states_interactive |>
```

```
  st_drop_geometry()|>
  count()
```

```
# A tibble: 1 x 1
      n
  <int>
1    51
```

Numeric variable (static plot):

```
ggplot(states_static, aes(x = long, y = lat, group = group)) +
  geom_polygon(aes(fill = carfatal07), color = "white", linewidth = 0.2) +
  coord_map() +
  theme_void() +
  scale_fill_viridis_c(na.value = "grey90") +
  labs(fill = "Car fatalities\n(2007)",
       title = "Motor vehicle fatality rates by state")
```



Numeric variable (interactive plot):

```r
# it's okay to skip a legend here
pal_num <- colorNumeric(
  palette = "viridis",
  domain = states_interactive$carfatal07,
  na.color = "grey90"
)

labels_num <- states_interactive |>
  mutate(
    label_text = str_c(
      "<strong>", name, "</strong><br/>",
      "Car fatalities (2007): ", round(carfatal07, 2)
    )
  ) |>
  pull(label_text) |>
  lapply(HTML)

leaflet(states_interactive) |>
  addTiles() |>
  setView(-96, 37.8, 4) |>
  addPolygons(
    fillColor = ~pal_num(carfatal07),
    weight = 1,
    color = "white",
    opacity = 1,
    fillOpacity = 0.8,
    label = labels_num,
    highlightOptions = highlightOptions(
      weight = 3,
      color = "#666",
      bringToFront = TRUE
    )
  )
```

Categorical variable (static plot):

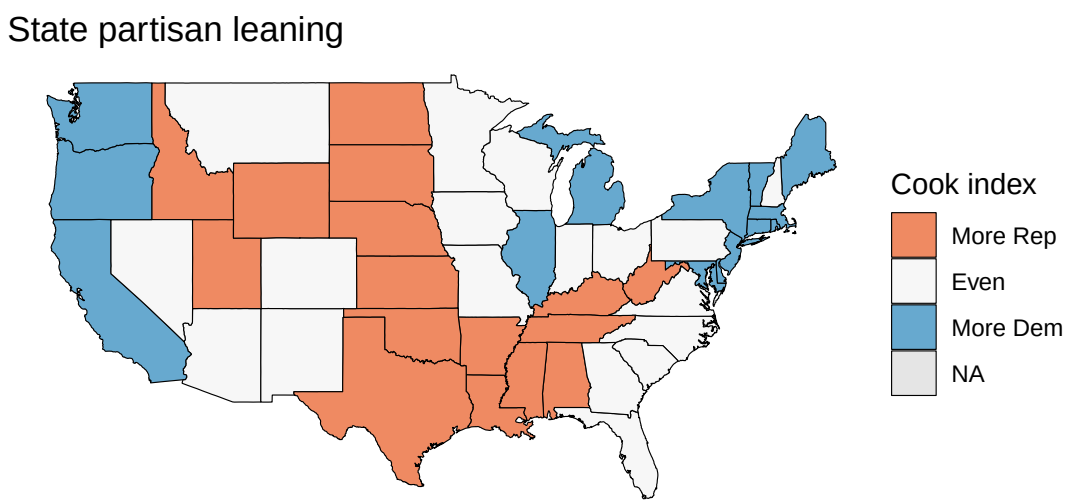
```r
# be really careful with matching color order to factor level order
states_interactive |>
  st_drop_geometry() |>
  count(cook_index3)
```

```
# A tibble: 4 x 2
```

```
  cook_index3     n
  <fct>       <int>
1 More Rep       16
2 Even           18
3 More Dem       15
4 <NA>            2
```

```r
# Make factor levels explicit so color order stays consistent
states_static <- states_static |>
  mutate(cook_index3 = factor(cook_index3))

ggplot(states_static, aes(x = long, y = lat, group = group)) +
  geom_polygon(aes(fill = cook_index3), color = "black", linewidth = 0.2) +
  coord_map() +
  theme_void() +
  scale_fill_brewer(palette = "RdBu", na.value = "grey90") +
  labs(fill = "Cook index",
       title = "State partisan leaning")
```



Categorical variable (interactive plot):

```r
# may use colorFactor() here
states_interactive <- states_interactive |>
  mutate(cook_index3 = factor(cook_index3))

pal_cat <- colorFactor(
  palette = "RdBu",
  domain = states_interactive$cook_index3,
  na.color = "grey90"
)

labels_cat <- states_interactive |>
  mutate(
    label_text = str_c(
      "<strong>", name, "</strong><br/>",
      "Cook index: ", cook_index3
    )
  ) |>
  pull(label_text) |>
  lapply(HTML)

leaflet(states_interactive) |>
  addTiles() |>
  setView(-96, 37.8, 4) |>
  addPolygons(
    fillColor = ~pal_cat(cook_index3),
    weight = 1,
    color = "white",
    opacity = 1,
    fillOpacity = 0.8,
    label = labels_cat,
    highlightOptions = highlightOptions(
      weight = 3,
      color = "#111",
      bringToFront = TRUE
    )
  ) |>
  addLegend(
    pal = pal_cat,
    values = ~cook_index3,
    title = "Cook index",
    position = "bottomright"
  )
```